Test-5 (Data structures)

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① A list contains the denominations as follows
D = [2000, 500, 200, 100, 50, 20, 10, 5]
Accept an amount from user and calculate how many minimum number of notes will be needed for that amount.

D = [2000, 500, 200, 100, 50, 20, 10, 5]

```
D = [2000, 500, 200, 100, 50, 20, 10, 5]
amount = int(input("Enter amount:"))
for note in D:
    count = amount // note
    if count > 0:
        print(note, ":", count)
        amount = amount % note
print("minimum no. of notes will be needed")
```

② A teacher come to class with a large box that has several coins Each coin has a number pointed on it. Before coming to class, she ensurred that All the numbers occurr an even no. of times. However, while coming to the class, one coin fell down and got lost She wants to find out the no. on the missing coin.

I/P:

~~Coins = [1, 2, 3, 4, 4, 5]~~

Coins = [1, 2, 3, 1, 2, 3, 4, 4, 5]

```
count = { }
for i in coins:
    if i in count:
        count [i] += 1
    else:
        count [i] = 1
for i in count:
    if count [i] % 2 != 0:
        print("missing coin no:", i)
```

③ A list contains sublist with emp information as follows;

Data = [[101, "seema", 45000], [340, "Rajan", 13000], [210, "Tannu", 14000], [320, "Suresh", 35000]]

Write a program to sort the list based on salary.

```
Data = [
    [101, "Seema", 45000],
    [340, "Rajani", 30000],
    [210, "Tannu", 14000],
    [320, "Suresh", 35000],
]
Data.sort(key = lambda x: x[2])
for emp in Data:
    print(emp)
```

4. There is a list with some numbers Create a new dictionary using list in such a way that key is no and value is freq of occurence of that no in list

[1, 3, 4, 1, 2, 3, 6, 7, 1, 2, 4]

{1: 3, 3: 2, 2: 2, 4: 2}

```
li = [1, 3, 4, 1, 2, 3, 6, 7, 1, 2, 4]
dict = { }
for i in li:
    if i in D:
        D[i] += 1
    else:
        D[i] = 1
print(D)
```

5. Python Program to find the union of two lists without using set concept

Date

```
L1 = [1, 2, 3, 4]
L2 = [3, 4, 5, 6]
Union = []
for i in L1:
    for i in L1:
        if i not in union:
            union.append(i)
    for i in L2:
        if i not in union:
            union.append(i)
print("Union of two lists:", union)
```